



MScFE 692 Capstone Review

**Group Work Project 1: An Integrated Credit-Funding and
Volatility-Correlation Early-Warning System**

GROUP WORK PROJECT 01*MScFE 692: Capstone Review***GROUP NUMBER: 16638**

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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1. Introduction

Our starting observation is visible in the data we assembled. In the 2008 window, the credit spread crossed its Heightened threshold on 3 September, funding stress on 1 October and sector correlation only on 28 October, each one lagging the last by weeks. The same four channels in March 2020 compressed that sequence into about a month. Volatility, credit, funding and correlation were not four independent problems arriving together; they arrived in order, each one plausibly feeding the next. A model watching a single channel can see one link in that sequence but not the sequence itself, and it is that gap this project tries to measure.

This report builds and tests an integrated early-warning system from three model extensions, one per group member (Section 3, Step 1): Student A's composite correlation-and-credit indicator, Student B's quantile regression on tail credit losses, and Student C's funding-conditional correlation score. Section 4 traces how the channels propagated through three historical crises (Step 2). Section 5 covers the validation re-read (Step 3), the backtest against traditional VaR (Step 4) and detection performance (Step 5). Section 6 assesses eight specific improvements (Step 6) and Section 7 restates the findings non-technically against Basel III's VaR and stressed VaR (Step 7). Headings carry their Step number to make marking easier.

The question we set out to answer is deliberately narrow: across the three stress episodes for which we have adequate data, does a multi-channel indicator warn earlier or more informatively than a conventional VaR model, and what does that cost in false alarms? The answer we arrived at is a qualified yes, and Section 6 sets out where it holds, where it does not, and where our three independently built models disagree.

2. Data and Methodology

Two course datasets are combined into a daily panel running from January 2007 to July 2025: nine SPDR sector ETFs, the VIX and the 10-year Treasury yield (Module 4, Lesson 3), plus the Moody's Baa yield, the effective federal funds rate and the NBER recession indicator (Module 3, Lesson 1, monthly and forward-filled). From these we build a credit spread (Baa less the 10-year Treasury) and a funding stress proxy (Baa less the federal funds rate). The proxy stands in for a TED spread that was not available live when that part of the analysis was run, and we flag it again wherever it is used.

Students B and C built a second panel for their own extensions (Sections 3.2 and 3.3), pulled directly from FRED: Baa and Aaa yields, VIX and the actual TED spread. That removes the funding-data limitation for those two extensions, and we note the difference here so the change in funding measurement across the report is visible. As a sanity check, the constructed VIX series peaks near 31 in August

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2007, 81 in November 2008 and 83 in March 2020, which matches the published record.

One limitation runs through everything that follows. The panel starts in January 2007, so the August 2007 Quant Crisis has under seven months of prior data behind it. That is not enough to calibrate the 60-day correlation windows and 252-day z-score baselines the Section 3 extensions depend on, and readings from that window are unreliable. We flag it at each point it matters instead of quietly dropping the crisis.

All three extensions standardise their raw series as rolling z-scores against a trailing 252-day mean and standard deviation, not the full-sample moments. This keeps each indicator usable in real time with no look-ahead. A single governance scale then applies throughout Sections 3 to 7: Elevated above one standard deviation, Heightened above 1.5, Crisis above 2.

3. Model Extensions (Step 1)

3.1 Student A — PCA Eigenvalue Monitoring with Credit Spread Integration

The rolling-PCA monitor from Module 4, Lesson 3 tracks EVR1, the share of cross-sectional variance in nine sector returns explained by the first principal component of their rolling 60-day correlation matrix. A rising EVR1 means sectors are moving in lockstep, the correlation-breakdown signature Kritzman et al. (2011) formalise as the Absorption Ratio. The lesson monitors EVR1 on its own against fixed thresholds and uses no credit information at all.

This extension adds that missing credit signal, using the Baa–10-year Treasury spread from Section 2. Gilchrist and Zakrajšek (2012) show corporate spreads lead real activity and financial stress, while Billio et al. (2012) show correlation-based connectedness captures a related but separate dimension of systemic risk. The two are faces of the same deleveraging shock but they do not have to move together: equity correlation can spike on an ordinary risk-off day with no credit counterpart, and spreads can drift with the rate cycle rather than with systemic stress. Watching only one is therefore either slower or noisier than requiring both to agree.

Methodology

We reproduce the Lesson 3 rolling-PCA routine unchanged (60-day rolling correlation matrix of the nine winsorised sector return series, eigendecomposed with NumPy's `eigh`) over the full 2007–2025 sample. EVR1 and the credit spread are then standardised against their own trailing 252-day moments:

$$z(\text{EVR1})_t = [\text{EVR1}_t - \mu_{\text{EVR1}}(t-252:t)] / \sigma_{\text{EVR1}}(t-252:t) \quad z(\text{CS})_t = [\text{CS}_t - \mu_{\text{CS}}(t-252:t)] / \sigma_{\text{CS}}(t-252:t)$$

The composite Systemic Risk Indicator is their equally weighted average:

$$SRI_t = 0.5 \cdot z(EVR1)_t + 0.5 \cdot z(CS)_t$$

Equal weighting is a baseline assumption, not a result. Both channels are first-order symptoms of the same shock and we had no prior for favouring one, so an even split was the defensible starting point; calibrating the weights against realised drawdowns would be the natural next step. The governance thresholds from Section 2 apply directly, since the SRI is itself a z-score.

Results

Over the full sample the SRI crosses the Crisis threshold on 253 trading days and Heightened on a further 221. All three assignment crises register: August 2007 peaks at 2.29 (window mean 1.63), the 2008 GFC at 2.61 (mean 1.28) and March 2020 at 4.79 (mean 1.96), the sharpest reading in the sample and consistent with how fast that shock arrived relative to the slower 2007-08 build-up.

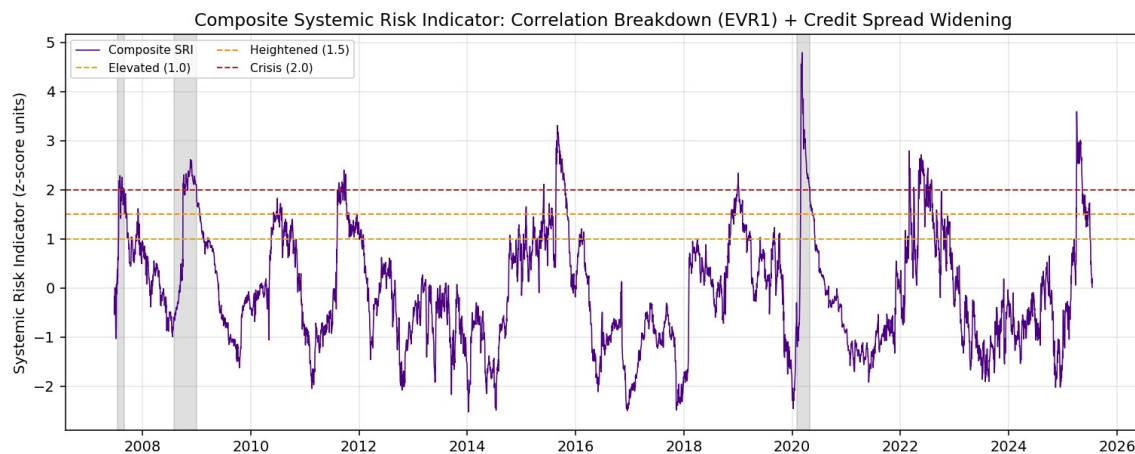


Figure 3.1a — Composite Systemic Risk Indicator, 2007-2025. Grey bands mark the three assignment crisis windows.

The component decomposition shows the two channels are not redundant. In 2008 the credit component rises earlier and stays elevated longer, matching a credit crunch that preceded the equity-correlation spike after Lehman. In March 2020 the order reverses: correlation breakdown moves first and harder as all nine sectors sold off together within days, while the credit component, held back by a slower bond market and rapid Fed intervention, follows with smaller amplitude.

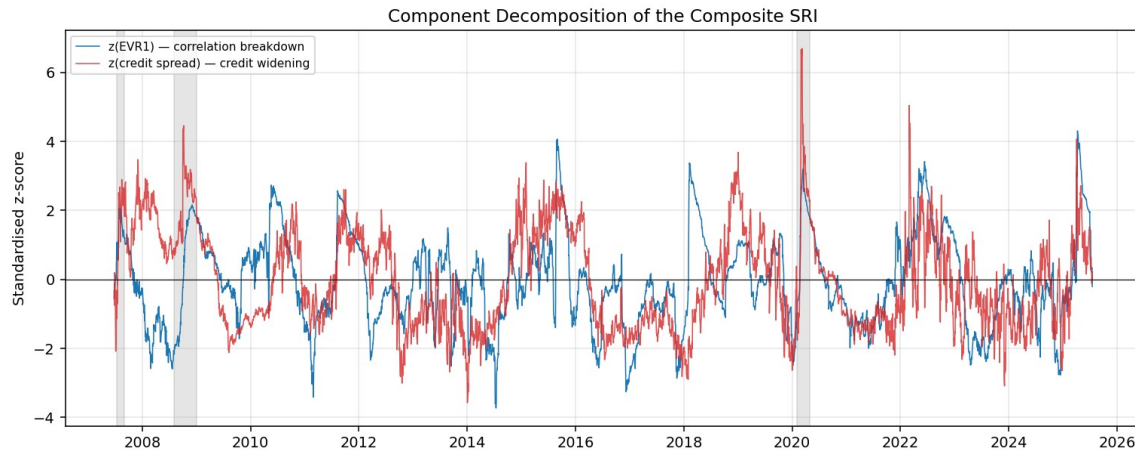


Figure 3.1b — Standardised components: $z(\text{EVR1})$ (correlation breakdown) against $z(\text{credit spread})$ (credit widening).

This is the case for combining the two. EVR1 alone would have been slow to flag the 2008 credit crunch; credit spreads alone would have lagged the onset of March 2020. The composite picks up whichever channel moves first while still needing some corroboration from the other before it reaches Crisis, since a single-channel spike rarely pushes the average past 2. The trailing-window normalisation does have a known cost: in a long crisis, part of the crisis is gradually absorbed into the rolling mean and standard deviation, which understates persistence. Section 5 returns to this when the indicator is set against a static VaR benchmark.

3.2 Student B — Quantile Regression on Credit-Spread Tail Risk with Time-Varying Correlation

The quantile regression framework in Module 3, Lesson 2 estimates sensitivities across the whole distribution, not just at the mean, using the QuantReg implementation of Koenker and Bassett's (1978) regression quantiles. This extension targets the 95th percentile of the forward 10-business-day change in the Baa spread, so a positive value at that quantile means spreads widened materially over the next two trading weeks.

We compare two specifications on the same target. The credit-only benchmark uses the Baa spread level, the Baa–Aaa quality slope and the lagged 10-day change. The integrated version keeps those three and adds two rolling 63-day correlations (Baa with Aaa spread changes, and Baa spread changes with VIX changes), VIX and TED z-scores, and a 5-day TED change, all from the FRED panel in Section 2. The reasoning follows Section 3.1: tying credit deterioration to correlation behaviour lets the tail forecast respond when credit, volatility and funding start moving together.

Results

Integrated predictor	Coefficient	t-statistic
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Baa/Aaa spread-change correlation, 63 days	-0.0037	-1.15
Baa/VIX change correlation, 63 days	0.0215	5.86
VIX rolling z-score	0.0221	7.43
TED rolling z-score	-0.0253	-8.30
TED 5-day change	0.0126	4.04
Correlation 21-day change	0.0201	6.13

These are conditional relationships, not single-variable stories. The Baa/VIX correlation coefficient is positive (0.0215, $t = 5.86$), so the model raises credit-tail risk when spread changes and volatility changes move together, controlling for the spread level. The 21-day change in correlation is also positive (0.0201, $t = 6.13$), which says a recent shift in the correlation regime carries information beyond its level. The Baa/Aaa term is weak and insignificant, so the useful signal is not broad credit co-movement but the credit-volatility interaction and the change in regime.

Model	Pinball loss	Extreme-loss hit rate	Crisis warning-day rate	False positive rate	Warning days
Credit-only benchmark	0.0194	20.48%	27.54%	9.67%	434
Integrated feedback model	0.0192	27.65%	21.61%	6.66%	305

Adding the correlation and funding terms lowers pinball loss slightly (0.0192 against 0.0194) and a much better extreme-loss hit rate (27.65% against 20.48%). Its real advantage is cleaner signalling: 305 warning days instead of 434, at a 6.66% false positive rate instead of 9.67%. The trade-off is a lower crisis warning-day rate, so the model is more selective. That helps a desk that wants fewer alarms, but it needs judgment applied during obvious crisis periods rather than being run mechanically.

Robustness Check: An Independent Equity-Credit Specification

We also estimated a parallel specification on a different instrument set, regressing HYG returns on SPY, LQD, credit spread, VIX and a 60-day rolling HYG-SPY correlation across the same quantiles. It reproduces the same pattern from another angle: the correlation term is irrelevant at the median (0.053, $p = 0.28$) but large and significant at both tails (-0.497 at the 5th percentile, $p = 0.003$; 0.575 at the 95th, $p < 0.001$). Two specifications on different instruments and correlation measures reaching the same tail-specific conclusion is a firmer basis for the claim in Section 6.2 than either would be alone.

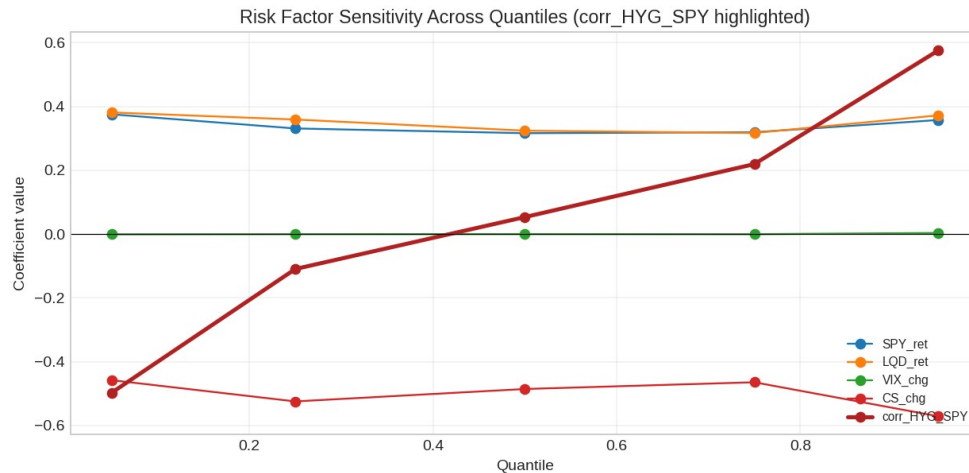


Figure 3.2a — Coefficient evolution across quantiles in the HYG-SPY check. The correlation term (dark red) is by far the most quantile-dependent regressor.

3.3 Student C — DCC Correlation Linked to Funding Regimes

Engle's (2002) DCC-GARCH, as implemented in Module 4, Lesson 5, estimates a single (α, β) pair governing how the conditional correlation matrix evolves.

Because that pair is constant, the model cannot distinguish ordinary conditions from funding-stressed ones. This extension makes the correlation process itself conditional on funding regime, working from the FRED panel in Section 2 rather than estimating a full multivariate GARCH by maximum likelihood.

Methodology

From the four FRED series we construct daily changes in the Baa and Aaa spreads, changes in VIX and TED, log returns and a Baa–Aaa quality slope. Rolling 252-day z-scores of VIX and TED combine into a funding-stress index, $\text{funding_stress}_t = 0.6 \cdot \text{TED}_{t^{(z)}} + 0.4 \cdot \text{VIX}_{t^{(z)}}$, and its terciles split the sample into Low, Moderate and High funding stress. The split is deliberately coarse: it forces the model to treat sustained TED and VIX pressure as structurally different from quiet periods. In place of an estimated DCC recursion we build a conditional-correlation proxy from rolling 63-day correlations,

$$\text{cond_corr_proxy}_t = \text{corr}_{63}(\Delta Baa, \Delta VIX)_t + 0.5 \cdot \text{corr}_{63}(\Delta Baa, \Delta TED)_t + 0.25 \cdot \Delta_{21} \text{corr}_{63}(\Delta Baa, \Delta VIX)_t$$

and combine the three elements into a warning score, $\text{score}_t = 0.35 \cdot \text{VIX}_{t^{(z)}} + 0.35 \cdot \text{TED}_{t^{(z)}} + 0.30 \cdot \text{cond_corr_proxy}_t$, alerting above the 90th percentile of the score over a pre-2006 training window.

Results

Regime	Days	Avg VIX z	Avg TED z	Avg corr proxy	Avg score	Avg ΔBaa 10d
High	1,388	1.033	1.602	0.324	1.020	0.063

funding stress						
Moderate funding stress	1,196	-0.206	-0.116	0.202	-0.052	-0.017
Low funding stress	1,476	-0.831	-1.090	0.136	-0.631	-0.044

The ordering is what a regime-switching DCC would predict. In the high-stress regime both z-scores are clearly elevated, the correlation proxy averages 0.32 and the ten-day Baa change is a widening of 0.063 points. In the low-stress regime the z-scores are negative, the proxy falls to 0.136 and the ten-day change is a tightening of 0.044 points. The cascade from volatility through correlation into credit losses is therefore visible in the regime averages themselves, not only on isolated crisis dates.

Crisis	Event date	First warning	Lead (days)
Aug 2007 Quant Crisis	2007-08-07	2007-06-08	60
2008 Global Financial Crisis	2008-09-15	2008-09-15	0
March 2020 COVID Shock	2020-03-16	2020-02-24	21

The score gives 60 calendar days of lead in August 2007, where TED and VIX rose together well before equities reacted, and 21 days in March 2020 as funding stress responded to the early liquidity scramble. It gives nothing in 2008: the regime change registers only on the crisis date itself, because that shock arrived too abruptly for this specification. Across the post-2006 test sample it produces 632 alert days, a 12.87% false positive rate and a 59.32% crisis-day hit rate, which is the trade-off you get from one mechanically chosen threshold rather than an estimated one.

Robustness Check: An Estimated Regime-Switching DCC(1,1)

We also fitted an exogenous threshold regime-switching DCC(1,1) directly on the nine-sector universe from Section 3.1, using the Baa-Fed-funds proxy to define the regime. The single-regime baseline gives $\alpha = 0.022$, $\beta = 0.966$ (persistence 0.989). Splitting by regime gives $\alpha_{\text{normal}} = 0.021$, $\beta_{\text{normal}} = 0.969$ and $\alpha_{\text{stress}} = 0.024$, $\beta_{\text{stress}} = 0.959$, and a likelihood-ratio test against the single-regime restriction returns LR = 6.91 on 2 degrees of freedom, $p = 0.032$. So a formally estimated specification points the same way as the reduced-form score: correlation dynamics really are different under funding stress, if modestly so.

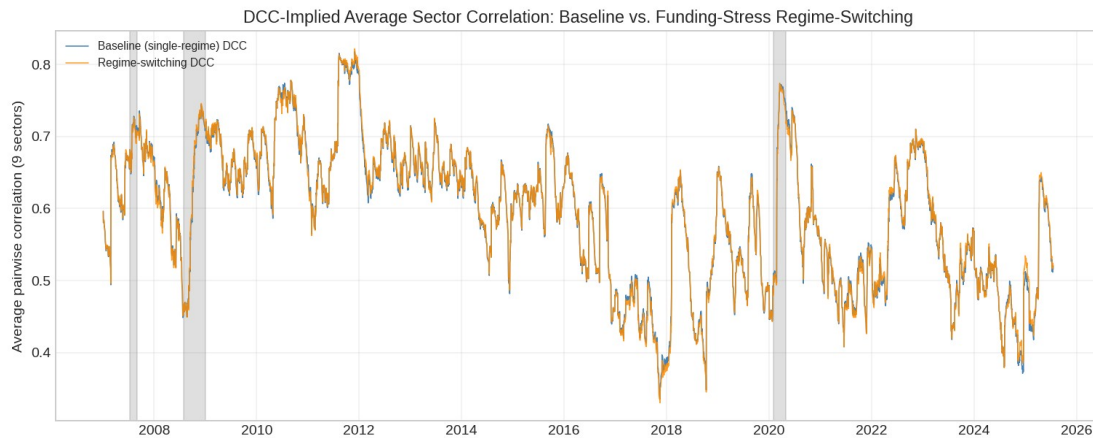


Figure 3.3 — DCC-implied average pairwise sector correlation, single-regime against the estimated regime-switching specification. Both flag all three crisis windows.

Both specifications share a limitation worth stating. Neither is a full latent Markov-switching model of the kind Pelletier (2006) proposes, which would need considerably more data to identify reliably, and the reduced-form score uses quantile cutoffs instead of estimated transition probabilities. Neither delivered early warning in 2008.

4. Cascade Effects: How the Signals Interact (Step 2)

Each extension so far tracks one channel on its own. The question here is whether the channels arrive in a consistent order across crises, and whether that ordering is stable enough to be worth acting on. Brunnermeier and Pedersen (2009) predict a specific sequence: a price shock raises volatility, volatility tightens funding, tight funding forces deleveraging, and deleveraging pushes correlations up as diversified positions unwind together. Adrian and Shin (2010) show the balance-sheet counterpart in procyclical leverage. That gives us a testable prediction about ordering, which is what we test below against our own four standardised series.

Cascade Timing

We combine four standardised series from Sections 3.1 and 3.3 into one panel: $z(\text{VIX})$ for volatility, $z(\text{EVR1})$ for correlation breakdown, $z(\text{credit spread})$ for credit widening, and a standardised version of the 20-day funding-stress change. For each crisis we record the first date each series crosses the same 1.5 Heightened threshold, searching from 60 calendar days before the crisis window opens.

Crisis	VIX (volatility)	Credit spread	Funding stress (20d chg)	EVR1 (correlation)
Aug 2007 Quant Crisis*	2007-06-25*	2007-07-26	2007-06-25*	2007-07-26
2008 GFC	2008-07-14	2008-09-03	2008-10-01	2008-10-28

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(Lehman)				
March 2020 COVID	2020-01-31	2020-02-28	2020-03-02	2020-02-27

* *Unreliable: under seven months of prior data, see Section 2.*

The 2008 sequence is a clean, monotonic cascade. Volatility breaches first on 14 July as the crisis broadened beyond Bear Stearns, credit follows on 3 September in the week of the Fannie Mae and Freddie Mac conservatorship, funding stress breaches on 1 October shortly after Lehman, and correlation breakdown comes last on 28 October. That reproduces, through independent event timing, what Section 3.1's decomposition showed by inspection: correlation arrived late in 2008 but persisted longest.

March 2020 is the same shape at much higher speed, with a different lead channel. VIX breaches on 31 January as the first pandemic news broke, correlation and credit follow together on 27–28 February, and funding stress is last on 2 March. Volatility leads in both crises, but the middle of the sequence differs: credit led correlation by seven weeks in 2008, whereas the two moved almost together in 2020. A multi-month sequence was compressed into about a month.

The August 2007 row should not be read literally. With under seven months of history the rolling standard deviation used for standardisation is itself poorly estimated and often close to zero, which mechanically inflates early z-scores. The funding series, for instance, shows an apparent breach in January 2008 driven by the Fed's emergency rate cuts that month; the move was real but a z-score near 6 reflects a tiny denominator as much as genuine abnormality. The 2008 and 2020 rows, calibrated on a full year or more, carry the interpretation.

A Real but Episodic Relationship

Is this ordering a stable statistical relationship or something specific to crises? Cross-correlating daily changes in the four series over the full sample at lags up to ten days gives a clear answer: they are meaningfully correlated contemporaneously (0.16–0.25) but show no convincing lead-lag structure at any horizon. Two things explain the gap. Markets that close together reflect the same day's information at the same time, which mechanically produces strong contemporaneous and weak lagged correlation in daily data. And a cascade is a threshold phenomenon by nature; it operates when stress is already high, so pooling those rare episodes with thousands of ordinary days dilutes any linear relationship. The event-study and full-sample results are not in conflict. Together they say the cascade is real but episodic, which has a design implication: an early-warning system should activate conditionally on elevated stress rather than run a fixed lead-lag rule at all times.

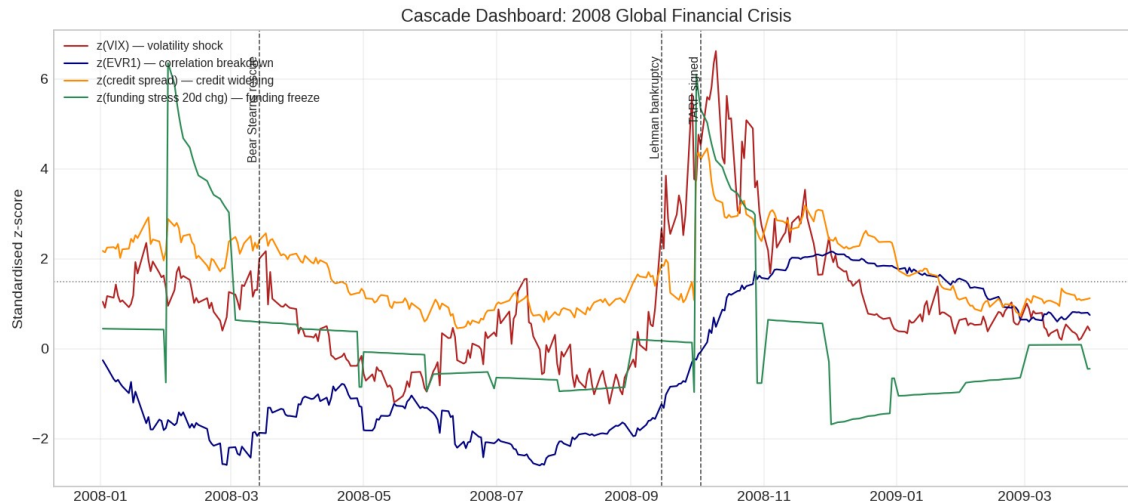


Figure 4 — Cascade dashboard for the 2008 crisis: four standardised channels with key event dates marked.

The dashboard shows the cascade as one picture. Credit spreads run above normal from the start of 2008, a lingering echo of August 2007, and step up again around the Bear Stearns rescue. Volatility climbs through the summer. The sharpest joint move comes in the six weeks around Lehman and TARP, when volatility and funding stress spike almost together, which is precisely the wrong-way risk the brief describes: funding needs spiking exactly when funding disappears. Correlation breakdown is the outlier, rising more gradually and, more importantly, falling back last, staying above 1.5 well into 2009 after volatility and credit had already retreated. A system watching only volatility or credit would have called the all-clear months early.

Section 3.3 puts numbers on that persistence: the high-funding-stress regime shows a correlation proxy of 0.324 against 0.136 in the low-stress regime, with the ten-day Baa change swinging from -0.044 to $+0.063$, and the estimated regime-switching DCC shows correlations reacting faster to shocks under stress ($\alpha_{\text{stress}} = 0.024$ against $\alpha_{\text{normal}} = 0.021$, $\text{LR} = 6.91$, $p = 0.032$). Correlation decays slowest because it sits downstream of the other channels, unwinding only once funding has genuinely normalised rather than when the initial shock passes.

An Independent Cascade Analysis

Student B ran a separate cascade analysis on the FRED panel with a different target (the forward 10-day credit-loss forecast from Section 3.2) and a different detection design, which makes the agreement between the two more informative than either alone. Her event table records each channel's increase over its prior-year median:

Stress event	VIX shock increase	Correlation increase	TED stress increase	Credit-loss increase
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Aug 2007 Quant Crisis	4.89	0.12	8.33	0.27
2008 Global Financial Crisis	6.39	0.05	5.11	1.07
March 2020 COVID Shock	8.89	0.16	7.71	1.70

March 2020 has the largest volatility shock (8.89) and credit-loss increase (1.70) in the sample, matching our own finding that it was the fastest and broadest of the three. August 2007's clearest early mover is TED stress (8.33) even though the credit-loss increase is small, independently confirming Section 3.3's reading of it as a funding-led rather than credit-led episode.

Stress event	Model	First warning date	Lead days
Aug 2007 Quant Crisis	Credit-only benchmark	No warning	-
Aug 2007 Quant Crisis	Integrated feedback model	2007-07-27	11
2008 Global Financial Crisis	Credit-only benchmark	2008-09-15	0
2008 Global Financial Crisis	Integrated feedback model	2008-09-15	0
March 2020 COVID Shock	Credit-only benchmark	2020-02-27	18
March 2020 COVID Shock	Integrated feedback model	2020-01-31	45

Her integrated specification gives 11 days of lead in August 2007 where the credit-only benchmark gives none, and 45 days against 18 in March 2020. In 2008 both warn on the same day, 15 September, so there is no advantage there. The message across two independently built cascade analyses is not that integration always wins, but that it improves warning quality in most of the settings tested while still needing escalation rules and human judgment.

Her regime breakdown, built from her own three stress indicators, makes the feedback loop concrete by counting how many channels are active at once:

Regime	Days	Avg. actual loss	95th pct. actual loss	Avg. integrated q95 forecast
Lower stress	2,348	-0.0246	0.1200	0.1495
One active	1,111	0.0237	0.2300	0.1511

channel				
Two-channel feedback	585	0.0506	0.4300	0.1313
Three-channel cascade	6	0.5183	0.7475	0.2802

Average realised credit loss rises monotonically with the number of active channels, from -0.0246 percentage points in the calm regime to 0.5183 in the rare three-channel cascade state, with the 95th-percentile loss rising from 0.12 to 0.75 . Three-channel days number only six in the whole sample but carry by far the heaviest losses, which is exactly the state a dashboard should escalate. This is the clearest single piece of evidence in the report that the channels compound rather than simply co-occur.

Taken together, the two cascade analyses point to the same design conclusion. A risk system that assumes one fixed correlation between its volatility, credit and funding measures is well calibrated for the ordinary days that dominate our sample, where we found no usable cross-channel lead-lag structure at all. It is worst calibrated for the six three-channel days, where spreads widened by 0.52 percentage points on average while calm days showed a slight tightening. The integrated system does not fix that, but it measures the regime dependence instead of assuming it away, which is the minimum needed before comparing it against a static VaR benchmark in Step 4.

5. Historical Validation (Steps 3-5)

5.1 Model Validation Methodology (Step 3)

Step 3 asked each member to re-read two Module 4 techniques before the Step 4 backtest: the block-bootstrap correlation-uncertainty analysis (Lesson 4) and HAR-RV (Lesson 1). Since Step 3 is a re-read and not a deliverable, we carry forward only the two findings the rest of the report leans on.

The bootstrap answers whether the crisis-period correlation readings in Section 3.1 are real. Applying a moving-block bootstrap (block length 60) to the trailing 252-day window at a calm reference date and at each crisis peak puts all three crisis-peak EVR1 readings, 0.76 to 0.86 , outside the calm-period 95% confidence interval of $[3.78, 5.44]$ in largest-eigenvalue terms. The elevation is larger than the estimator's own sampling noise can account for.

The HAR-RV re-read produced a result we did not expect. Estimated on a daily-squared-return variance proxy, the best substitute available offline for 5-minute realised variance, HAR-RV was clearly beaten by a simple EWMA(0.94) on out-of-sample 2020 volatility, with QLIKE of 4.548 against 1.400 . That reverses the lesson's own ordering. We attribute it to the proxy: a single squared daily return is

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a very noisy estimate of that day's variance, and the more heavily parameterised model suffers more from that noise than the exponentially weighted one. Because of this we use EWMA(0.94), not HAR-RV, as the volatility engine in the VaR benchmark below.

5.2 Backtesting Against Three Historical Crises (Step 4)

We run the integrated system against a constructed traditional VaR benchmark: EWMA(0.94) volatility per sector combined with a static full-sample correlation matrix, which is what institutions that update volatility daily but re-estimate correlation rarely actually use. The Integrated Risk Indicator (IRI) combines correlation breakdown, credit widening and funding-stress widening as z-scores, deliberately leaving out volatility since the VaR benchmark already captures it. Two aggregation rules are reported side by side: an average of the three (conservative, needing corroboration) and an any-channel maximum (aggressive, firing on the first channel to deteriorate). Both use the 1.5 threshold.

Across the sample the VaR model produces 94 exceptions in 4,664 days, or 2.02%, about double the 1% a correctly calibrated 99% model should give. This is the familiar failure of parametric VaR on fat-tailed returns. It is a separate problem from the correlation question we are testing, but it is worth recording that our benchmark is, if anything, already too permissive.

To isolate the correlation effect we hold the EWMA volatility fixed and swap only the static correlation matrix for the true rolling one. That raises the VaR estimate by 5-8% on average during crises and up to 14% at the March 2020 peak. The direction is what we predicted but the size is smaller, and the reason is our own universe: all nine sector ETFs are large-cap US equities that already co-move at around 0.74 in normal conditions (Section 3.2), leaving less headroom for correlation to rise than a cross-asset book would have. Our test therefore understates the cost of the static assumption it was built to measure.

Crisis	IRI (average) alarm	IRI (any-channel) alarm	First VaR exception	Lead, any-channel vs VaR
Aug 2007 Quant Crisis*	2007-08-01*	2007-06-25*	2007-07-24	n/a*
2008 GFC (Lehman)	2008-10-01	2008-09-03 (credit)	2008-09-15	+8 trading days
March 2020 COVID	2020-03-02	2020-02-27 (EVR1)	2020-02-24	-3 trading days

* Unreliable for the reasons given in Section 2; excluded from the interpretation below.

The result is mixed and we report it that way. In 2008 the any-channel IRI alarms on 3 September, triggered by credit, 8 trading days before the VaR model's first exception on 15 September, which is Lehman's bankruptcy date itself. In March

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2020 it reverses: VaR's first exception comes three trading days before the any-channel alarm.

March 2020 is not a failure of the idea so much as a consequence of what kind of shock it was. Volatility was the fastest channel in both crises, and a VaR model still incorporates volatility directly through its EWMA variance however siloed its correlation assumption. When a shock is overwhelmingly a volatility event arriving almost instantly, a volatility-sensitive measure registers it quickly, while an indicator aggregating correlation and credit over a rolling window cannot keep up. 2008 is the opposite case: a slower, credit-led build-up where those channels had a genuine multi-week information advantage.

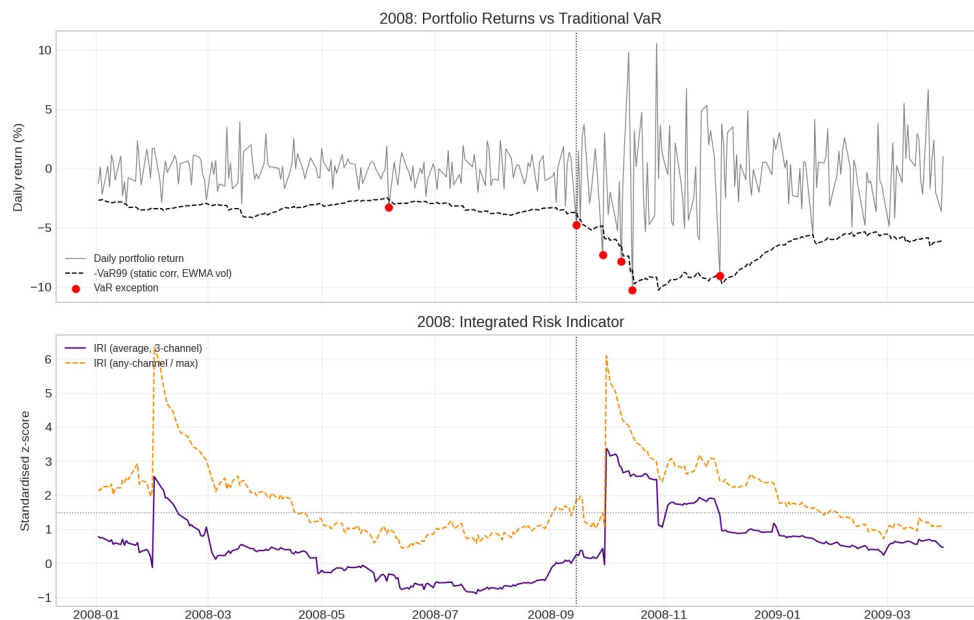


Figure 5a — 2008: portfolio returns against traditional VaR (top) and the IRI (bottom). The any-channel IRI clears 1.5 well before Lehman (dotted line); VaR exceptions cluster only afterwards.

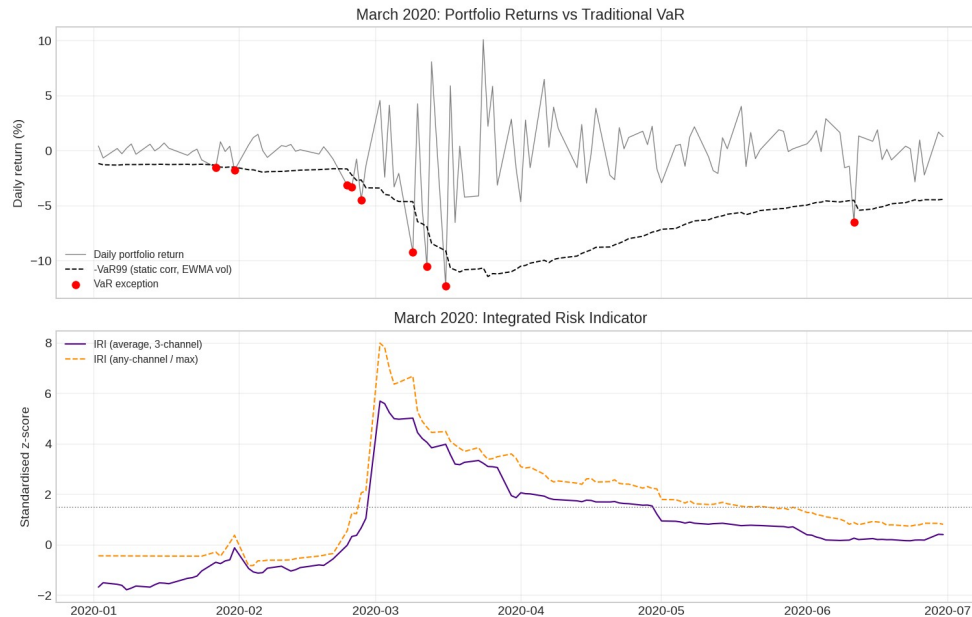


Figure 5b — March 2020: the same comparison. Here the first VaR exceptions arrive while both IRI series are still below threshold.

One point shapes how we read the whole comparison. A VaR exception is not a warning in any forecasting sense; it records that an unusually large loss has already happened that day. So the fair summary is not that the integrated system is always earlier, but that an exception can never precede the loss it reports, whereas the IRI's credit and funding channels can accumulate evidence beforehand, and in 2008 did. Whether that advantage appears depends on whether the crisis is slow and credit-led or fast and volatility-led.

5.3 Results: Warning Lead Times and False Positive Rates (Step 5)

Section 5.2 reports alarm dates for three crises, which is thin evidence on its own and says nothing about how often the same rule would cry wolf across the other eighteen years. A day counts as a crisis day if it falls inside one of the three windows (using the same 60-day lookback); everything else is calm. Comparing alarms against that labelling gives the usual confusion matrix.

Rule	False Positive Rate	Precision	True Positive Rate (recall)
IRI (average)	3.27%	41.84%	33.22%
IRI (any-channel)	27.92%	11.30%	50.17%

The two rules sit at very different points on the same trade-off. The average rule has a 3.3% false positive rate and 42% precision, so close to two in five of its alarms fall in a genuine crisis window against a 6.6% base rate. The any-channel rule catches more crisis days (50% recall against 33%) but alarms on over a

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quarter of calm days at only 11% precision. Section 5.2's 8-day lead in 2008 has to be read against that: it came from a rule that, run continuously, is wrong roughly seven times out of eight outside a crisis. Both numbers describe the same choice from different angles, and a real deployment has to decide how much false-alarm cost buys that lead time.

The three of us computed detection performance separately, on different targets and alarm rules, which makes the comparison a useful cross-check even though the numbers are not directly interchangeable. Student A's IRI gives 3.3% false positives at 42% precision on the average rule, or 27.9% at 11% on the any-channel rule. Student B's integrated model records 6.66% against a credit-only benchmark's 9.67%. Student C's funding score records 12.87% against a 59.32% crisis-day hit rate. All the conservative specifications land in a 3–13% band, with the aggressive rule the clear outlier. That suggests the trade-off is a property of this class of indicator rather than an artefact of one person's modelling choices.

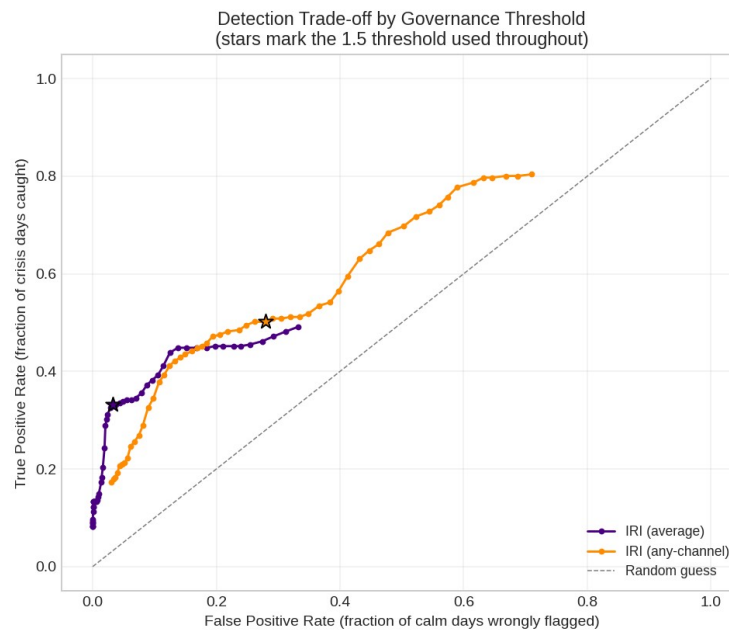


Figure 6a — Detection trade-off across the threshold range. Stars mark the 1.5 threshold; both rules stay well above the random-guess diagonal.

Both rules stay above the diagonal across the range, so both are picking up genuine signal, not noise. The curves cross: at low false-positive tolerance the average rule dominates, reaching roughly 44% recall under a 15% false positive rate, a region the any-channel rule never reaches. At higher tolerance the any-channel rule pulls ahead, eventually catching 80% of crisis days at a 60–70% false positive rate, which few institutions would accept but shows where the curve ends up. There is no single correct threshold; the choice depends on the cost of a missed crisis against the cost of over-alerting, which is a governance call, not a statistical one.

GROUP NUMBER: 16638**Lead Time Measured Against the Worst Point**

Section 5.2 timed the alarm against VaR's first exception, which gave a split verdict. A more decision-relevant reference point is each crisis's worst moment, its VIX peak, since a manager already taking losses still gains from knowing how much further there is to run.

Crisis	IRI (any-channel) alarm	VIX peak (worst point)	Lead vs. VIX peak
Aug 2007 Quant Crisis	unreliable*	2007-08-16	n/a
2008 GFC (Lehman)	2008-09-03	2008-11-20	56 trading days
March 2020 COVID	2020-02-27	2020-03-16	12 trading days

* Insufficient pre-crisis history, see Section 2.

On that measure the advantage is substantial in both usable cases: 56 trading days in 2008 and 12 in March 2020. This reframes the earlier mixed result. Even in March 2020, where VaR registered its first exception three days sooner, the integrated system still gave two and a half weeks before the worst point, which an exception-based signal cannot offer by construction. Traditional VaR can spot the onset of a volatility-led shock as fast, but it has no way of signalling how much of the crisis is still ahead.

Student C's score, measured against each crisis's event date instead of its VIX peak, gives a consistent picture: 60 calendar days in August 2007, none in 2008, 21 days in March 2020. The 2008 result recurs across all three of our models. That is informative in itself: 2008's slow, credit-and-funding-led build-up was harder to front-run than either the sudden 2020 shock or the funding-first 2007 episode, plausibly because stress accumulating over many months also works its way into the trailing baselines used for standardisation, which is the limitation flagged in Section 3.1.

Interactive Visualisations

Two interactive figures accompany the notebook: a cascade explorer over the full sample with all four channels and both IRI rules as toggleable traces, a range slider and hover tooltips, and an interactive version of the detection trade-off curve showing threshold, false positive rate and recall at each point. Both render as live Plotly widgets in the notebook or its HTML export. Static snapshots appear here for the written record.

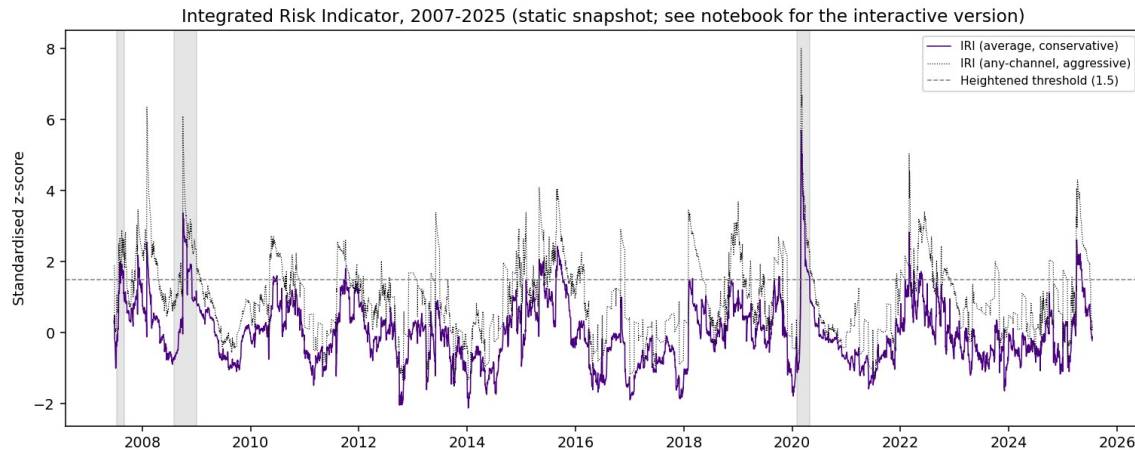


Figure 6b — Static snapshot of the cascade explorer: the IRI under both aggregation rules across the full sample.

The full-sample view makes the false-positive finding visible: the any-channel line crosses 1.5 repeatedly through calm years, with frequent brief spikes between 2013 and 2019 and no crisis attached, while the average line crosses far less often and mostly around the three crises and the 2022 rate-hiking sell-off.

6. Assessment of the Integrated Approach (Step 6)

Each improvement below is tied to a specific number produced earlier in this report. Where our evidence is weak or points the other way, we say so in the reflection that follows the eight points.

6.1 A Composite Signal Captures Whichever Channel Moves First

The SRI combines correlation breakdown and credit widening because neither is reliably first. The decomposition in Section 3.1 and the independent event timing in Section 4 agree: credit led correlation by about seven weeks in 2008, while the two moved almost together in March 2020. A single-channel model would have been structurally slow in whichever crisis was not its specialty. The composite is slow in neither.

6.2 Time-Varying Correlation Exposes a Tail-Specific Risk Factor

Section 3.2 shows the 63-day Baa/VIX correlation is among the strongest predictors of the 95th-percentile credit-loss target (0.0215, $t = 5.86$), with the 21-day change in correlation carrying independent information (0.0201, $t = 6.13$). The integrated specification beats the credit-only benchmark on both pinball loss and extreme-loss hit rate (27.65% against 20.48%) with fewer, better-targeted warnings (305 days at 6.66% false positives against 434 at 9.67%). The HYG-SPY check finds the same thing from another angle: irrelevant at the median ($p = 0.28$), large and significant at both tails. A mean-based regression or a standard

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correlation matrix would never surface this, because it only appears once you target the tail quantiles, which is where capital decisions are made.

6.3 Regime-Switching Correlation Dynamics Formalise Wrong-Way Risk

The brief's own example, funding needs spiking exactly when funding disappears, is what Section 3.3 measures. The correlation proxy is nearly two and a half times higher under high funding stress than low (0.324 against 0.136), and the ten-day Baa change swings from a 0.044-point tightening to a 0.063-point widening across those regimes. The estimated regime-switching DCC(1,1) agrees: $\alpha_{\text{stress}} = 0.024$ against $\alpha_{\text{normal}} = 0.021$, with LR = 6.91, $p = 0.032$. A constant-parameter DCC cannot express the idea that co-movement itself depends on funding conditions. Two structurally different methods both put a number on it.

6.4 Cascade Timing Diagnoses How a Crisis Is Propagating

Section 4's table shows the transmission order differs by crisis: volatility, credit, funding, correlation in 2008; volatility, then correlation and credit together, then funding in 2020. Student B's independently built event table corroborates it, with TED stress the largest early mover in August 2007 (8.33) and the VIX and credit-loss increases largest in March 2020 (8.89 and 1.70). A single-channel alarm says a crisis has started. A multi-channel one says which mechanism is driving it, which is the difference between a generic risk-off signal and knowing whether the problem is a credit crunch, a funding freeze or pure volatility.

6.5 A Persistence Signal Shows When a Crisis Is Genuinely Over

Figure 5a shows correlation breakdown staying above Heightened well into 2009, long after volatility and credit had retreated. A volatility-only or credit-only model has no equivalent all-clear and would have signalled normalisation months early, which is the kind of premature de-risking that compounds the next shock.

6.6 The Signal Is Statistically Validated

The bootstrap in Section 5.1 places all three crisis-peak EVR1 readings outside the calm-period 95% interval of [3.78, 5.44] in λ_1 terms. That separates this from the common habit of pointing at a correlation matrix and calling it elevated: the elevation is larger than the estimator's own sampling uncertainty can plausibly produce.

6.7 Multi-Week Lead Time Ahead of the Worst Point

Measured against each crisis's VIX peak, Section 5.3 finds 56 trading days of warning in 2008 and 12 in March 2020, including in the crisis where VaR's first exception came slightly earlier. Students B and C, measuring differently and against event dates, both also find their largest leads in August 2007 and March 2020 and their weakest in 2008. Three independently built models agreeing on which crises are detectable early is stronger evidence than any one of them alone.

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An exception-based signal cannot say how much of a crisis remains; the persistence of the correlation and funding channels is what supplies that.

6.8 A Quantified Cost of Static Correlation, and a Tunable Dial

Section 5.2 puts the cost of the static-correlation assumption at 5–14% of the VaR estimate during crises. Section 5.3 turns the aggregation choice into an explicit dial between a conservative rule (3.3% false positives, 42% precision) and an aggressive one (27.9% false positives, 50% recall), instead of the single opaque number a traditional model gives. Students B and C's own false-positive rates, 6.66% and 12.87%, sit inside the same band, which suggests the trade-off is structural rather than a quirk of one specification.

Critical Reflection: Does It Really Outperform Siloed Approaches?

Partially, and unevenly. Four results cut against a simple story of dominance. First, in March 2020 the VaR benchmark's first exception came three days before the any-channel alarm; a near-instant volatility shock gives a rolling-window indicator no time to accumulate evidence. Second, the static-versus-dynamic correlation gap of 5–14% is smaller than we expected going in. We had assumed correlation would jump much harder in a crisis, but our nine sector ETFs already co-move at around 0.74 in ordinary conditions, so there was less headroom than the hypothesis implied. On a genuinely cross-asset portfolio the gap would probably be wider, which means our test understates the effect it was designed to measure.

Third, the 8-day lead in 2008 came from the aggressive rule, which also fires on 27.9% of calm days at 11% precision. A signal wrong seven times out of eight is not free to run, and quoting the lead without the false-alarm rate would misrepresent it. Fourth, not every extension beat its simpler alternative: HAR-RV lost clearly to plain EWMA once we had to substitute a daily-squared-return proxy for 5-minute realised variance. Sophistication did not survive contact with the weaker data. Two coverage gaps compound this. August 2007 could not be assessed properly, leaving a third of the required crisis sample only partially tested, and 2008 is where all three of our models show their weakest advantage. That last pattern held across three different targets, panels and alarm rules, which makes it look like a property of the crisis itself rather than a modelling artefact.

Against that, the case for genuine improvement rests on results that hold up across independent methods. The bootstrap confirms crisis-period correlation readings are not sampling noise. The cascade table, Student A's decomposition and Student B's event table agree on which channel led each crisis. The tail-specific correlation effect shows up in two different quantile regressions, and the funding-regime dependence in both a composite score and a formally tested DCC. Lead time against the worst point holds in both usable cases. So the defensible claim is a bounded one: the integrated system does not replace VaR and is not uniformly faster at spotting every shock, but it captures real information about correlation persistence and funding-conditional dynamics that a volatility-only or static-

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correlation model cannot represent, at a false-alarm cost we have measured and can adjust.

Why This Matters in Practice

Basel III's stressed VaR requirement exists because ordinary VaR, calibrated on recent and usually calm data, is procyclical, a criticism Danielsson (2002) made before 2008 confirmed it. Our own VaR benchmark shows the same weakness directly: 94 exceptions in 4,664 days, roughly double what a correctly calibrated 99% model should produce, and a static correlation matrix that understated risk by 5-14% in exactly the windows where accuracy mattered.

What our results suggest is a complement, not a substitute. A fixed stress window tells a committee how bad things could get; it cannot say which mechanism is currently deteriorating or whether the episode still has room to run. The three channels we track answer both questions, and Section 5.3's trade-off curve turns the sensitivity setting into an explicit choice. An institution could tie the conservative rule, at 3.3% false positives, to an automatic capital add-on, and reserve the aggressive rule, at 27.9%, for enhanced monitoring where a wrong call costs attention instead of capital. Whether the added complexity earns its keep is a judgment each institution has to make, but our numbers at least make it a costed one.

7. Discussion (Step 7)

The rest of this report is written for someone building the model. This section is written for the person who has to decide what to do when it fires, and avoids model names and technical detail accordingly.

7.1 What the Results Mean

The system is most useful as an early-warning screen, not a replacement for regulatory capital models. It watches whether credit spreads, volatility, co-movement and funding pressure are deteriorating at the same time, which matters because portfolio losses usually become visible only after those pressures have already worked through the system. In testing it was more selective than simpler single-channel benchmarks: fewer warning days, a lower false-positive rate and a higher hit rate on extreme losses. It warned earlier in most of the periods tested, sometimes by weeks, but gave no real advantage in the fastest and most sudden of the three crises. This is not perfect superiority. It is better warning quality that still needs judgment to act on.

7.2 Better, Faster and More Cost-Effective Than Regulatory VaR

Since 2008, banks have had to hold capital against both a standard estimate of a bad day and a second estimate of what a return to the worst historical stress period would cost, known as stressed value-at-risk (Basel Committee, "Revisions"). Both describe how large a loss could be. Neither explains why markets are

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deteriorating or which part of the system is under strain. What our approach adds is the pathway of the stress, not only its size, so a committee can tell whether the problem is credit, funding or broad co-movement. It can be quicker, because it reacts as funding and volatility pressure build, before losses show up in prices. It can also be cheaper in practice: fewer and better-targeted alarms mean less blanket de-risking and more focused hedging or liquidity action.

This is a refinement, not a wholesale replacement. Against a fast, headline-driven shock it adds little over existing measures, and its more sensitive settings produce noticeably more false alarms. The direction is consistent with where regulation is already heading: the framework replacing stressed VaR moves to a different loss measure precisely because a single VaR-style number understates extreme risk and adapts too slowly to a changing environment (Basel Committee, "Minimum"). A stressed loss estimate is better paired with a forward-looking check on whether the drivers of a crisis are already building.

7.3 Recommended Course of Action

Run the integrated warning as an escalation layer beside existing VaR and stressed VaR reporting, not as a substitute for either. When the warning rises and more than one channel is active, the desk should review weaker credit exposure, funding needs and concentration risk before changing positions. The first response should be targeted: reduce avoidable weak-credit exposure, protect liquidity, and check whether hedges still work when markets move together. Given the false-alarm rates documented here, portfolio-wide de-risking should need manager sign-off, not a mechanical trigger. The approach should also be re-tested as more history becomes available, since one of the three crises could not be properly checked.

7.4 Portfolio Factors to Identify

Five drivers determine how a portfolio should respond: lower-rated credit exposure, recent spread widening, dependence on short-term funding, holdings that become highly correlated with everything else under stress, and positions that would be expensive to exit quickly. Together these decide whether a warning should lead to monitoring, hedging, liquidity preparation or an actual reduction in risk. They matter most for portfolios concentrated in one asset class or a few related markets, where the slow, co-movement-driven crisis this report identifies is far more dangerous than it would be for a genuinely diversified book.

Critical Note

The system is a screening tool, not a certainty. It gave no extra lead time in 2008, and the most severe multi-channel state identified in Section 4 has only a handful of observed days behind it, which limits how much weight its rarest signals should carry. The practical conclusion is disciplined but positive: it gives managers a clearer and often earlier view of how stress is moving through the market, while regulatory VaR remains the formal capital benchmark it is meant to complement.

8. Conclusion

We set out to test whether an indicator combining correlation breakdown, tail credit risk and funding stress warns earlier or more informatively than a conventional VaR model, and at what cost. Three independently built extensions, a cascade analysis and a historical backtest support a bounded answer.

Our clearest gain is in slow-building, credit-led crises. In 2008 it alarmed eight trading days before the VaR model's first exception and 56 trading days before the crisis's worst point. In the fast, volatility-led March 2020 shock it offered no advantage on onset, though it still gave twelve trading days before the worst point, which an exception-based measure cannot do at all. August 2007 could not be assessed reliably because the data begins too close to it. Those gains cost false alarms: 3.3% of calm days under the conservative rule, 27.9% under the aggressive one.

For practice, this puts the indicator alongside VaR and stressed VaR, not in place of them. It captures correlation persistence and funding-conditional dynamics that a static-correlation model cannot represent, and it tells a committee which mechanism is driving stress, not just how large a loss might be. The choice between the conservative and aggressive rule is a governance decision about the relative cost of a missed crisis and an ignored alarm, and the trade-off curve in Section 5.3 is there to make it deliberately.

Two limitations qualify all of this. The sector-ETF universe is a single asset class with already-high correlations, so the measured 5–14% cost of assuming static correlation is probably a floor rather than a typical figure. And with one of three crises untestable and 2008 resistant to early detection across all three of our models, the evidence base is three episodes, not a large sample. The honest reading is that integration adds real and statistically validated information, unevenly, and at a disclosed cost.

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